SUMIT BANDYOPADHYAY

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Engineering leader with over 21+ years of experience in driving large-scale software and product development for automotive and emerging technologies. Experienced in leading global teams of 130+, managing and delivering end-to-end programs aligned with business and stakeholder objectives. Skilled in project governance, risk management, and ASPICE-compliant execution across the full V-model lifecycle. Strong record of improving delivery efficiency, reducing cost overruns, and boosting customer satisfaction. Adept at aligning cross-functional teams and fostering innovation in Al driven solutions with domain strengths in automotive systems and embedded software.

PROFESSIONAL EXPERIENCE

Founder

Jul 2024 – Dec 2025

Aspire AI Solutions

Established AspireAI Solutions as an early-stage venture to explore AI-driven solutions across automotive and enterprise domains.

Focused on proof-of-concept development in computer vision, multi-agent systems.

Designed and validated small-scale AI/ML workflows (TensorFlow, PyTorch, Hugging Face) with integration into MLflow and Kubeflow for lifecycle management.

Experimented with autonomous agent frameworks (LangChain, LangGraph, CrewAI, AutoGen) to orchestrate diagnostic intelligence and workflow automation.

Explored deployment of Retrieval-Augmented Generation (RAG) pipelines and cloud-native AI prototypes on AWS, Azure, and GCP.

Actively building an ecosystem of early prototypes and MVPs, with a roadmap toward scalable domain-specific AI products.

Senior SW Department Manager

Jun 2017 – Jul 2024

Valeo India Pvt Ltd

Built and scaled a high-performing software department from the ground up, growing to 150+ engineers at the offshore development center and delivering embedded solutions for 48V hybrid systems including iBSGs, DC-DC converters, inverters, and onboard chargers.

Spearheaded strategic department growth by enhancing execution efficiency, elevating team maturity, and implementing quality-focused delivery practices.

Collaborated with cross-functional teams (Development, Testing, VCore, Quality, Functional Safety & Cybersecurity) across department to deliver end-to-end automotive software solutions.

Successfully aligned development roadmaps with global stakeholders across France, Germany, China, and Egypt, enabling end-to-end autonomy across the software lifecycle.

Oversaw daily operations, resource planning, and team support while ensuring smooth execution across all delivery tracks.

Contributed to the development of software solutions for Valeo's Software Defined Vehicle (SDV) initiatives, enhancing ECU intelligence, fault detection, and system reliability.

Monitored ASPICE L2 dashboards and milestone progress, ensuring process compliance and audit readiness across all projects.

Tracked and improved key project health metric-reducing schedule slippage by 35% and cost overrun by 20%, with insights reported regularly to the steering committee.

Maintained 98% resource utilization by optimizing billable vs. non-billable allocations within a $1M annual operating budget.

Achieved 100% staffing for forecasted headcount within six weeks through targeted recruitment and streamlined onboarding.

Elevated customer satisfaction by 10% YoY, supported by a 4.6/5 CSAT score and consistent delivery of ASPICE L2-compliant milestones.

Reduced tool duplication by 40% and achieved 12% cost savings through rationalization and reuse initiatives.

Championed R&D efficiency and innovation initiatives, driving idea generation and successfully filing patents for intellectual property.

Mentored 20+ engineers into domain experts under Valeo's technical leadership track, fostering a strong internal talent pipeline.

EDUCATION

Deakin University

Master's in Data Science

2025

University of Texas

Post Graduate Program in AIML

2023

ITM University, Mumbai

Executive Master's in Business Administration

2014

Mumbai University

Bachelor's in Engineering, Electronics

2002

Managed annual appraisals, compensation cycles, and attrition strategy—achieving a low 8% attrition rate in a competitive talent market.

Technical Project Manager

Jun 2014 – May 2017

HCL Tech Ltd

Led ECU development for off-highway vehicles with a global 30+ member team, delivering 12+ milestones aligned with ISO 26262 and ASPICE standards.

Improved defect closure by 40% and reduced project risk by 30% through effective Dev-Test coordination and proactive tracking.

Handled customer billing, provided revenue forecasts, responded to 15+ RFPs, and secured $500K+ in new business.

Delivered successful AUTOSAR-based IO Master ECU integration and J1939 diagnostics, mentoring 20 engineers and improving skills by 25%.

Lead Engineer

Dec 2007 – Jun 2014

Lear India Engineering Center

Led design and development of Car Access System modules on CAN/LIN within AUTOSAR, covering requirement analysis, implementation, diagnostics, and V-model lifecycle.

Served as AUTOSAR Architect for a European client, delivering design solutions, BSW/MCAL/OS integration, and application consulting.

Managed full V-model lifecycle, ensuring 95% test coverage and zero critical post-release bugs.

Built AUTOSAR tools and test frameworks, improving build and testing efficiency by 40–50%.

Software Tech Lead

Jul 2005 – Dec 2007

Embin Technologies Pvt Ltd

Embedded Trainee Engineer

Jul 2004 – Mar 2005

Micromax Informatics Pvt Ltd

Software Engineer

Apr 2003 – May 2004

Unique Diagnostic Solutions